

DevSaaS Portfolio — Documentazione Codice

Architettura completa: React, TypeScript, Firebase Realtime Database, Auth & API Service

1. Struttura del Progetto

```
my-saas-project/  
  .env.local  
  vite.config.ts  
  index.html  
  src/  
    main.tsx  
    App.tsx  
    index.css  
    components/  
      NavigationBar.tsx  
      Footer.tsx  
      ProtectedRoute.tsx  
    context/  
      AuthContext.tsx  
    firebase/  
      firebaseConfig.ts  
    pages/  
      Home.tsx  
      Login.tsx  
      Register.tsx  
      Dashboard.tsx  
    services/  
      apiService.ts
```

2. Configurazione Ambiente, Firebase & API Services

FILE: vite.config.ts

```
import { defineConfig } from 'vite'  
import react from '@vitejs/plugin-react'  
  
export default defineConfig({  
  plugins: [react()],  
})
```

FILE: .env.local

```
VITE_FIREBASE_API_KEY=AIzaSyCIEAE45f7jel3404e8NrQeoveWz0p-uA  
VITE_FIREBASE_AUTH_DOMAIN=my-saas-principale.firebaseio.com  
VITE_FIREBASE_DATABASE_URL=https://my-saas-principale-default-rtdb.europe-west1.firebaseio.com  
VITE_FIREBASE_PROJECT_ID=my-saas-principale  
VITE_FIREBASE_STORAGE_BUCKET=my-saas-principale.firebaseio.com  
VITE_FIREBASE_MESSAGING_SENDER_ID=113575428119  
VITE_FIREBASE_APP_ID=1:113575428119:web:930eb179528306f9c8dc52
```

FILE: src/firebase/firebaseConfig.ts

```
import { initializeApp } from "firebase/app";
import { getAuth, Auth } from "firebase/auth";
import { getDatabase, Database } from "firebase/database";

const firebaseConfig = {
  apiKey: import.meta.env.VITE_FIREBASE_API_KEY as string,
  authDomain: import.meta.env.VITE_FIREBASE_AUTH_DOMAIN as string,
  databaseURL: import.meta.env.VITE_FIREBASE_DATABASE_URL as string,
  projectId: import.meta.env.VITE_FIREBASE_PROJECT_ID as string,
  storageBucket: import.meta.env.VITE_FIREBASE_STORAGE_BUCKET as string,
  messagingSenderId: import.meta.env.VITE_FIREBASE_MESSAGING_SENDER_ID as string,
  appId: import.meta.env.VITE_FIREBASE_APP_ID as string,
};

const app = initializeApp(firebaseConfig);
export const auth: Auth = getAuth(app);
export const db: Database = getDatabase(app);
```

FILE: src/services/apiService.ts

```
export interface DeveloperTask {
  id: number;
  title: string;
  completed: boolean;
}

export const getDeveloperStats = async (): Promise<DeveloperTask[]> => {
  return [
    { id: 1, title: 'Configurazione ambiente Firebase', completed: true },
    { id: 2, title: 'Implementazione Auth Context', completed: true },
    { id: 3, title: 'Ottimizzazione componenti UI React', completed: false },
  ];
};
```

FILE: Regole di Sicurezza Realtime Database

```
{
  "rules": {
    "orders": {
      ".read": "auth != null",
      "$orderId": {
        ".write": "auth != null && (!data.exists() || data.child('userId').val() === auth.uid) && newData.child('userId').val() === auth.uid"
      }
    }
  }
}
```

3. Context Autenticazione & Componenti Layout

FILE: src/context/AuthContext.tsx

```
import React, { createContext, useContext, useEffect, useState } from 'react';
import {
  User,
  createUserWithEmailAndPassword,
  signInWithEmailAndPassword,
  signOut,
  onAuthStateChanged
} from 'firebase/auth';
import { auth } from '../firebase/firebaseConfig';

interface AuthContextType {
  currentUser: User | null;
  signup: (email: string, pass: string) => Promise<any>;
  login: (email: string, pass: string) => Promise<any>;
  logout: () => Promise<void>;
}

const AuthContext = createContext<AuthContextType | undefined>(undefined);

export const useAuth = () => {
  const context = useContext(AuthContext);
  if (!context) throw new Error("useAuth deve essere usato all'interno di AuthProvider");
  return context;
};

export const AuthProvider: React.FC<{ children: React.ReactNode }> = ({ children }) => {
  const [currentUser, setCurrentUser] = useState<User | null>(null);
  const [loading, setLoading] = useState<boolean>(true);

  const signup = (email: string, pass: string) => createUserWithEmailAndPassword(auth, email, pass);
  const login = (email: string, pass: string) => signInWithEmailAndPassword(auth, email, pass);
  const logout = () => signOut(auth);

  useEffect(() => {
    const unsubscribe = onAuthStateChanged(auth, (user) => {
      setCurrentUser(user);
      setLoading(false);
    });
    return unsubscribe;
  }, []);

  return (
    <AuthContext.Provider value={{ currentUser, signup, login, logout }}>
      {!loading && children}
    </AuthContext.Provider>
  );
};
```

FILE: src/components/NavigationBar.tsx

```
import React from 'react';
import { Navbar, Nav, Container, Button } from 'react-bootstrap';
import { Link, useNavigate } from 'react-router-dom';
import { useAuth } from '../context/AuthContext';

const NavigationBar: React.FC = () => {
  const auth = useAuth();
  const { currentUser, logout } = auth as any;
  const navigate = useNavigate();

  const handleLogout = async () => {
    try {
      await logout();
      navigate('/login');
    } catch (error) {
      console.error("Errore durante il logout", error);
    }
  };

  return (
    <Navbar bg="dark" variant="dark" expand="lg" className="shadow-sm">
      <Container>
        <Navbar.Brand as={Link} to="/" className="fw-bold text-primary">
          <i className="bi bi-code-slash me-2"></i>DevSaaS
        </Navbar.Brand>
        <Navbar.Toggle aria-controls="basic-navbar-nav" />
        <Navbar.Collapse id="basic-navbar-nav">
          <Nav className="me-auto">
            <Nav.Link as={Link} to="/">Home</Nav.Link>
            {currentUser && <Nav.Link as={Link} to="/dashboard">Dashboard</Nav.Link>}
          </Nav>
          <Nav className="align-items-center gap-2">
            {currentUser ? (
              <>
                <span className="text-light small me-2">
                  <i className="bi bi-person-circle me-1"></i>{currentUser.email}
                </span>
                <Button variant="outline-light" size="sm" onClick={handleLogout}>Logout</Button>
              </>
            ) : (
              <>
                <Nav.Link as={Link} to="/login">Login</Nav.Link>
                <Button as={Link as any} to="/register" variant="primary" size="sm">Registrati</Button>
              </>
            )}
          </Nav>
        </Navbar.Collapse>
      </Container>
    </Navbar>
  );
};

export default NavigationBar;
```

FILE: src/components/Footer.tsx

```
import React from 'react';
import { Container } from 'react-bootstrap';

const Footer: React.FC = () => {
  return (
    <footer className="bg-dark text-white text-center py-3 mt-auto w-100">
      <Container>
        <p className="mb-0 small">&copy; {new Date().getFullYear()} DevSaaS Portfolio. Tutti i diritti riserv
ati.</p>
      </Container>
    </footer>
  );
};

export default Footer;
```

FILE: src/components/ProtectedRoute.tsx

```
import React from 'react';
import { Navigate } from 'react-router-dom';
import { useAuth } from '../context/AuthContext';

interface ProtectedRouteProps {
  children: React.ReactNode;
}

const ProtectedRoute: React.FC<ProtectedRouteProps> = ({ children }) => {
  const { currentUser } = useAuth();

  if (!currentUser) {
    return <Navigate to="/login" replace />;
  }

  return <>{children}</>;
};

export default ProtectedRoute;
```

4. Pagine dell'Applicazione

FILE: src/pages/Home.tsx

```
import React from 'react';
import { Container, Row, Col, Card, Button } from 'react-bootstrap';
import { useNavigate } from 'react-router-dom';
import { useAuth } from '../context/AuthContext';

const services = [
  { id: 'saas-starter', title: 'SaaS Starter Kit', price: '€299', desc: 'Template React + Firebase con autenticazione e database pronti.' },
  { id: 'fullstack-app', title: 'Applicazione Full-Stack Custom', price: '€899', desc: 'Sviluppo completo su misura con API dedicate e Dashboard.' },
  { id: 'cyber-audit', title: 'Cybersecurity & Code Audit', price: '€499', desc: 'Analisi delle vulnerabilità del codice e delle regole Firebase.' }
];

const Home: React.FC = () => {
  const { currentUser } = useAuth();
  const navigate = useNavigate();

  const handleSelectService = (projectName: string) => {
    if (currentUser) {
      navigate('/dashboard', { state: { selectedProject: projectName } });
    } else {
      navigate('/login', { state: { feedback: 'Effettua il login per ordinare un servizio.' } });
    }
  };

  return (
    <Container className="py-5">
      <div className="text-center mb-5">
        <h1 className="fw-bold display-5">Soluzioni Web & Development</h1>
        <p className="text-muted lead">Scegli il servizio ideale per il tuo prossimo progetto digitale</p>
      </div>
      <Row className="g-4">
        {services.map((item) => (
          <Col md={4} key={item.id}>
            <Card className="h-100 shadow-sm border-0 text-center p-3">
              <Card.Body className="d-flex flex-column">
                <Card.Title className="fw-bold h4 mb-3">{item.title}</Card.Title>
                <Card.Text className="text-muted flex-grow-1">{item.desc}</Card.Text>
                <div className="my-3">
                  <span className="display-6 fw-bold text-primary">{item.price}</span>
                </div>
                <Button variant="primary" size="lg" onClick={() => handleSelectService(item.title)}>
                  Richiedi Ora
                </Button>
              </Card.Body>
            </Card>
          </Col>
        ))}
      </Row>
    </Container>
  );
};

export default Home;
```

FILE: src/pages/Login.tsx

```

import React, { useState } from 'react';
import { Container, Card, Form, Button, Alert } from 'react-bootstrap';
import { useAuth } from '../context/AuthContext';
import { useNavigate, useLocation, Link } from 'react-router-dom';

const Login: React.FC = () => {
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const [error, setError] = useState('');
  const [loading, setLoading] = useState(false);

  const { login } = useAuth();
  const navigate = useNavigate();
  const location = useLocation();
  const feedback = location.state?.feedback;

  const handleSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    try {
      setError('');
      setLoading(true);
      await login(email, password);
      navigate('/dashboard');
    } catch (err: any) {
      setError("Credenziali non valide o errore di connessione.");
    } finally {
      setLoading(false);
    }
  };

  return (
    <Container className="d-flex align-items-center justify-content-center py-5" style={{ minHeight: '80vh' }}>
      <Card className="shadow border-0 p-4" style={{ maxWidth: '400px', width: '100%' }}>
        <Card.Body>
          <h3 className="text-center fw-bold mb-4">Accedi</h3>
          {feedback && <Alert variant="info">{feedback}</Alert>}
          {error && <Alert variant="danger">{error}</Alert>}
          <Form onSubmit={handleSubmit}>
            <Form.Group className="mb-3" controlId="formEmail">
              <Form.Label>Email</Form.Label>
              <Form.Control type="email" required value={email} onChange={(e) => setEmail(e.target.value)} placeholder="nome@esempio.com" />
            </Form.Group>
            <Form.Group className="mb-4" controlId="formPassword">
              <Form.Label>Password</Form.Label>
              <Form.Control type="password" required value={password} onChange={(e) => setPassword(e.target.value)} placeholder="Password" />
            </Form.Group>
            <Button variant="primary" type="submit" className="w-100 mb-3" disabled={loading}>
              {loading ? 'Accesso in corso...' : 'Accedi'}
            </Button>
          </Form>
          <div className="text-center small">
            Non hai un account? <Link to="/register">Registrati</Link>
          </div>
        </Card.Body>
      </Card>
    </Container>
  );
};

```

```
export default Login;
```


FILE: src/pages/Register.tsx

```

import React, { useState } from 'react';
import { Container, Card, Form, Button, Alert } from 'react-bootstrap';
import { useAuth } from '../context/AuthContext';
import { useNavigate, Link } from 'react-router-dom';

const Register: React.FC = () => {
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const [confirmPassword, setConfirmPassword] = useState('');
  const [error, setError] = useState('');
  const [loading, setLoading] = useState(false);

  const { signup } = useAuth();
  const navigate = useNavigate();

  const handleSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    if (password !== confirmPassword) {
      return setError("Le password non coincidono.");
    }
    try {
      setError('');
      setLoading(true);
      await signup(email, password);
      navigate('/dashboard');
    } catch (err: any) {
      setError("Impossibile creare l'account. Password troppo corta (min 6 caratteri) o email già in uso.");
    } finally {
      setLoading(false);
    }
  };

  return (
    <Container className="d-flex align-items-center justify-content-center py-5" style={{ minHeight: '80vh' }}>
      <Card className="shadow border-0 p-4" style={{ maxWidth: '400px', width: '100%' }}>
        <Card.Body>
          <h3 className="text-center fw-bold mb-4">Registrazione</h3>
          {error && <Alert variant="danger">{error}</Alert>}
          <Form onSubmit={handleSubmit}>
            <Form.Group className="mb-3" controlId="regEmail">
              <Form.Label>Email</Form.Label>
              <Form.Control type="email" required value={email} onChange={(e) => setEmail(e.target.value)} />
            </Form.Group>
            <Form.Group className="mb-3" controlId="regPassword">
              <Form.Label>Password</Form.Label>
              <Form.Control type="password" required value={password} onChange={(e) => setPassword(e.target.v
            </Form.Group>
            <Form.Group className="mb-4" controlId="regConfirmPassword">
              <Form.Label>Conferma Password</Form.Label>
              <Form.Control type="password" required value={confirmPassword} onChange={(e) => setConfirmPassw
            </Form.Group>
            <Button variant="success" type="submit" className="w-100 mb-3" disabled={loading}>
              {loading ? 'Creazione in corso...' : 'Crea Account'}
            </Button>
          </Form>
          <div className="text-center small">
            Hai già un account? <Link to="/login">Accedi</Link>
          </div>
        </Card.Body>
      </Container>
    )
  );
};

```

```
    </Card>
  </Container>
);
};

export default Register;
```

FILE: src/pages/Dashboard.tsx

```

import React, { useState, useEffect } from 'react';
import { Container, Row, Col, Card, Alert, Button, ListGroup, Spinner } from 'react-bootstrap';
import { useAuth } from '../context/AuthContext';
import { useLocation, useNavigate } from 'react-router-dom';
import { db } from '../firebase/firebaseConfig';
import { ref, push, set, onValue } from 'firebase/database';
import { getDeveloperStats, DeveloperTask } from '../services/apiService';

interface Order {
  id: string;
  projectName: string;
  status: string;
  userId: string;
}

const Dashboard: React.FC = () => {
  const { currentUser, logout } = useAuth();
  const location = useLocation();
  const navigate = useNavigate();

  const [feedback, setFeedback] = useState<string>(location.state?.feedback || '');
  const [orders, setOrders] = useState<Order[]>([]);
  const [apiData, setApiData] = useState<DeveloperTask[]>([]);
  const [loadingApi, setLoadingApi] = useState<boolean>(true);

  useEffect(() => {
    const selectedProject = location.state?.selectedProject;
    if (selectedProject && currentUser) {
      const saveOrder = async () => {
        try {
          const ordersRef = ref(db, 'orders');
          const newOrderRef = push(ordersRef);
          await set(newOrderRef, {
            userId: currentUser.uid,
            userEmail: currentUser.email,
            projectName: selectedProject,
            status: 'In Elaborazione',
            createdAt: new Date().toISOString()
          });
          setFeedback(`Richiesta inviata con successo per: ${selectedProject}!`);
        } catch (error) {
          console.error("Errore nel salvataggio dell'ordine", error);
        }
      };
      saveOrder();
    }
  }, [location.state, currentUser]);

  useEffect(() => {
    if (!currentUser) return;
    const ordersRef = ref(db, 'orders');
    const unsubscribe = onValue(ordersRef, (snapshot) => {
      const data = snapshot.val();
      if (data) {
        const userOrders: Order[] = Object.keys(data)
          .map(key => ({ id: key, ...data[key] }))
          .filter(order => order.userId === currentUser.uid);
        setOrders(userOrders);
      } else {
        setOrders([]);
      }
    });
  });
};

```

```

    return () => unsubscribe();
  }, [currentUser]);

useEffect(() => {
  getDeveloperStats()
    .then(data => {
      setApiData(data);
      setLoadingApi(false);
    })
    .catch(() => setLoadingApi(false));
}, []);

const handleLogout = async () => {
  await logout();
  navigate('/login');
};

return (
  <Container className="py-5 mx-auto">
    <div className="d-flex justify-content-between align-items-center mb-4">
      <div>
        <h2 className="fw-bold mb-1">Dashboard Utente</h2>
        <p className="text-muted mb-0">Benvenuto/a, <strong>{currentUser?.email}</strong></p>
      </div>
      <Button variant="outline-danger" onClick={handleLogout}>Logout</Button>
    </div>

    {feedback && (
      <Alert variant="success" dismissible onClose={() => setFeedback('')} className="mb-4">
        {feedback}
      </Alert>
    )}

    <Row className="g-4">
      <Col lg={6} md={12}>
        <Card className="shadow-sm border-0 h-100">
          <Card.Header className="bg-primary text-white fw-bold py-3">I Tuoi Servizi Ordinati</Card.Header>
          <Card.Body>
            {orders.length === 0 ? (
              <p className="text-muted mb-0">Non hai ancora richiesto nessun servizio.</p>
            ) : (
              <ListGroup variant="flush">
                {orders.map((order) => (
                  <ListGroup.Item key={order.id} className="d-flex justify-content-between align-items-center py-3">
                    <span className="fw-semibold">{order.projectName}</span>
                    <span className="badge bg-warning text-dark px-3 py-2">{order.status}</span>
                  </ListGroup.Item>
                ))}
              </ListGroup>
            )}
          </Card.Body>
        </Card>
      </Col>

      <Col lg={6} md={12}>
        <Card className="shadow-sm border-0 h-100">
          <Card.Header className="bg-dark text-white fw-bold py-3">Live API Feed (Task in Lavorazione)</Card.Header>
          <Card.Body>
            {loadingApi ? (
              <div className="text-center py-4">
                <Spinner animation="border" size="sm" className="me-2" /> Caricamento API...
              </div>
            ) : (
              <div>
                <h3>Live API Feed</h3>
                <p>Visualizza i dati in tempo reale dalla nostra API. I dati sono aggiornati automaticamente ogni 30 secondi.</p>
                <table>
                  <thead>
                    <tr>
                      <th>ID</th>
                      <th>Progetto</th>
                      <th>Stato</th>
                      <th>Ultimo Aggiornamento</th>
                    </tr>
                  </thead>
                  <tbody>
                    <tr>
                      <td>1</td>
                      <td>Progetto Alpha</td>
                      <td>In corso</td>
                      <td>2023-10-27 10:30</td>
                    </tr>
                    <tr>
                      <td>2</td>
                      <td>Progetto Beta</td>
                      <td>Completato</td>
                      <td>2023-10-26 15:45</td>
                    </tr>
                    <tr>
                      <td>3</td>
                      <td>Progetto Gamma</td>
                      <td>In attesa</td>
                      <td>2023-10-27 09:15</td>
                    </tr>
                    <tr>
                      <td>4</td>
                      <td>Progetto Delta</td>
                      <td>In corso</td>
                      <td>2023-10-27 11:00</td>
                    </tr>
                    <tr>
                      <td>5</td>
                      <td>Progetto Epsilon</td>
                      <td>In attesa</td>
                      <td>2023-10-27 08:30</td>
                    </tr>
                    <tr>
                      <td>6</td>
                      <td>Progetto Zeta</td>
                      <td>In corso</td>
                      <td>2023-10-27 12:15</td>
                    </tr>
                    <tr>
                      <td>7</td>
                      <td>Progetto Eta</td>
                      <td>In attesa</td>
                      <td>2023-10-27 07:45</td>
                    </tr>
                    <tr>
                      <td>8</td>
                      <td>Progetto Theta</td>
                      <td>In corso</td>
                      <td>2023-10-27 13:00</td>
                    </tr>
                    <tr>
                      <td>9</td>
                      <td>Progetto Iota</td>
                      <td>In attesa</td>
                      <td>2023-10-27 06:30</td>
                    </tr>
                    <tr>
                      <td>10</td>
                      <td>Progetto Kappa</td>
                      <td>In corso</td>
                      <td>2023-10-27 14:00</td>
                    </tr>
                    <tr>
                      <td>11</td>
                      <td>Progetto Lambda</td>
                      <td>In attesa</td>
                      <td>2023-10-27 05:15</td>
                    </tr>
                    <tr>
                      <td>12</td>
                      <td>Progetto Mu</td>
                      <td>In corso</td>
                      <td>2023-10-27 15:30</td>
                    </tr>
                    <tr>
                      <td>13</td>
                      <td>Progetto Nu</td>
                      <td>In attesa</td>
                      <td>2023-10-27 04:45</td>
                    </tr>
                    <tr>
                      <td>14</td>
                      <td>Progetto Xi</td>
                      <td>In corso</td>
                      <td>2023-10-27 16:00</td>
                    </tr>
                    <tr>
                      <td>15</td>
                      <td>Progetto Omicron</td>
                      <td>In attesa</td>
                      <td>2023-10-27 03:30</td>
                    </tr>
                    <tr>
                      <td>16</td>
                      <td>Progetto Pi</td>
                      <td>In corso</td>
                      <td>2023-10-27 17:15</td>
                    </tr>
                    <tr>
                      <td>17</td>
                      <td>Progetto Rho</td>
                      <td>In attesa</td>
                      <td>2023-10-27 02:45</td>
                    </tr>
                    <tr>
                      <td>18</td>
                      <td>Progetto Sigma</td>
                      <td>In corso</td>
                      <td>2023-10-27 18:30</td>
                    </tr>
                    <tr>
                      <td>19</td>
                      <td>Progetto Tau</td>
                      <td>In attesa</td>
                      <td>2023-10-27 01:15</td>
                    </tr>
                    <tr>
                      <td>20</td>
                      <td>Progetto Upsilon</td>
                      <td>In corso</td>
                      <td>2023-10-27 19:45</td>
                    </tr>
                    <tr>
                      <td>21</td>
                      <td>Progetto Phi</td>
                      <td>In attesa</td>
                      <td>2023-10-27 00:30</td>
                    </tr>
                    <tr>
                      <td>22</td>
                      <td>Progetto Chi</td>
                      <td>In corso</td>
                      <td>2023-10-27 20:15</td>
                    </tr>
                    <tr>
                      <td>23</td>
                      <td>Progetto Psi</td>
                      <td>In attesa</td>
                      <td>2023-10-27 00:00</td>
                    </tr>
                    <tr>
                      <td>24</td>
                      <td>Progetto Omega</td>
                      <td>In corso</td>
                      <td>2023-10-27 21:00</td>
                    </tr>
                    <tr>
                      <td>25</td>
                      <td>Progetto Zeta</td>
                      <td>In attesa</td>
                      <td>2023-10-27 00:45</td>
                    </tr>
                    <tr>
                      <td>26</td>
                      <td>Progetto Eta</td>
                      <td>In corso</td>
                      <td>2023-10-27 22:00</td>
                    </tr>
                    <tr>
                      <td>27</td>
                      <td>Progetto Theta</td>
                      <td>In attesa</td>
                      <td>2023-10-27 01:30</td>
                    </tr>
                    <tr>
                      <td>28</td>
                      <td>Progetto Iota</td>
                      <td>In corso</td>
                      <td>2023-10-27 23:15</td>
                    </tr>
                    <tr>
                      <td>29</td>
                      <td>Progetto Kappa</td>
                      <td>In attesa</td>
                      <td>2023-10-27 02:00</td>
                    </tr>
                    <tr>
                      <td>30</td>
                      <td>Progetto Lambda</td>
                      <td>In corso</td>
                      <td>2023-10-27 24:00</td>
                    </tr>
                  </tbody>
                </table>
              </div>
            )}
          </Card.Body>
        </Card>
      </Col>
    </Row>
  </Container>
);

```

```
        </div>
      ) : (
        <ListGroup variant="flush">
          {apiData.map((task) => (
            <ListGroup.Item key={task.id} className="small py-2 border-0">
              <i className={`bi ${task.completed ? 'bi-check-circle-fill text-success' : 'bi-clock te
xt-secondary'} me-2`} ></i>
              {task.title}
            </ListGroup.Item>
          )))
        </ListGroup>
      )}
    </Card.Body>
  </Card>
</Col>
</Row>
</Container>
);
};

export default Dashboard;
```

5. Entry Point Applicazione

FILE: src/App.tsx

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import { AuthProvider } from '../context/AuthContext';
import NavigationBar from '../components/NavigationBar';
import Footer from '../components/Footer';
import ProtectedRoute from '../components/ProtectedRoute';
import Home from '../pages/Home';
import Login from '../pages/Login';
import Register from '../pages/Register';
import Dashboard from '../pages/Dashboard';

function App() {
  return (
    <AuthProvider>
      <Router>
        <div className="d-flex flex-column min-vh-100 w-100 bg-light">
          <NavigationBar />
          <main className="flex-grow-1 w-100">
            <Routes>
              <Route path="/" element={<Home />} />
              <Route path="/login" element={<Login />} />
              <Route path="/register" element={<Register />} />
              <Route path="/dashboard" element={
                <ProtectedRoute>
                  <Dashboard />
                </ProtectedRoute>
              } />
            </Routes>
          </main>
          <Footer />
        </div>
      </Router>
    </AuthProvider>
  );
}

export default App;
```

FILE: src/main.tsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import 'bootstrap/dist/css/bootstrap.min.css';
import 'bootstrap-icons/font/bootstrap-icons.css';
import './index.css';

ReactDOM.createRoot(document.getElementById('root') as HTMLElement).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```